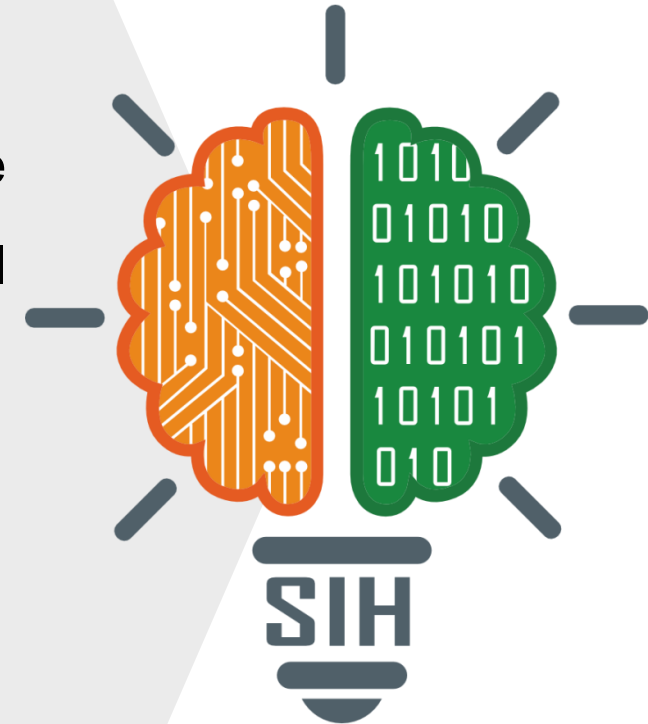


FoodLoop AI

- **Problem Statement ID** – 26234
- **Problem Statement Title**- AI-Powered Smart Food Waste Reduction and Sustainable Redistribution Ecosystem for Institutional Kitchens and Food Processing Units
- **Theme**- Agriculture, FoodTech & Rural Development
- **PS Category**- Software
- **Team ID**-
- **Team Name (Registered on portal)** - HackPulses



FoodLoop AI

PROBLEM

- Excess food is difficult to predict
- Overproduction leads to unnecessary food waste
- Suitable recipients are difficult to identify quickly
- Limited real-time visibility of surplus, safety & redistribution

SOLUTION

- AI-powered food surplus prediction
- Smart Matching Engine connects surplus food with nearby eligible recipients
- Food Safety Intelligence checks time, storage & redistribution eligibility
- Waste Analytics Dashboard tracks waste, savings & redistribution impact

INNOVATION

- Predicts how much food will become surplus before it becomes waste
- Combines AI Prediction + Smart Matching + Food Safety + Real-time Tracking
- Dynamic Rescue Radius prioritizes recipients based on distance & remaining food life
- AI Menu & Production Optimization helps kitchens prevent recurring waste
- Continuous Predict → Prevent → Redistribute → Analyze → Improve loop

TECHNOLOGIES

- **Frontend** – React.js
- **Backend** – Node.js + Express.js
- **Database** – MongoDB
- **AI/ML** – Python + Machine Learning Models
- **LLM** – LLM + Embeddings + NLP
- **Computer Vision** – OpenCV / Vision Model
- **Maps & Location** – Google Maps API / OpenStreetMap
- **Authentication** – JWT
- **Notifications** – Firebase Cloud Messaging

IMPLEMENTATION FLOW

Kitchen Data → AI Demand Prediction → Surplus Detection → Food Safety
Check → Smart Recipient Matching → Pickup Coordination → Impact
Analytics

FOOD DATA PROCESSING

- Menu & production data collection
- Historical consumption analysis
- Attendance / demand analysis
- Surplus quantity estimation
- Food category classification
- Preparation & expiry-time tracking

SMART REDISTRIBUTION ENGINE

- Identifies available surplus food
- Finds nearby eligible recipients
- Matches based on **quantity + distance + food type + urgency**
- Dynamic rescue radius based on remaining safe-use time
- Automated pickup coordination & notifications

FEASIBILITY AND VIABILITY

FEASIBILITY

- **Web-based** and accessible through standard devices
- **Modular & scalable** software architecture
- Can integrate with **existing kitchen management systems**
- Supports multiple institutions — **colleges, hospitals, hotels & factories**
- Cloud-based deployment enables **centralized monitoring and scaling**
- Uses existing **AI, maps, notification & database technologies**

CHALLENGES / RISKS

- Inaccurate **food surplus prediction**
- Poor-quality or incomplete kitchen data
- **Food safety** and redistribution-time constraints
- Recipient availability may change in real time

STRATEGIES

- **Historical-data + real-time feedback** for continuous prediction improvement
- Data validation and **admin verification**
- Rule-based **food safety validation** alongside AI
- Real-time recipient availability & capacity updates
- **AI-based smart matching** using distance, quantity, food type & urgency

IMPACT AND BENEFITS

FOR KITCHEN / FOOD PROCESSING UNITS

- Predict daily food demand & surplus
- Reduce overproduction and unnecessary waste
- Monitor food surplus in real time
- Optimize procurement and production
- Track cost savings and waste patterns

FOR NGOS / RECIPIENTS

- Receive real-time surplus food alerts
- Smart matching based on location & capacity
- View food quantity, type & safe-use window
- Faster pickup coordination
- Track meals received and beneficiaries served

OVERALL IMPACT

Traditional: Overproduction → Food Surplus → Delay → Food Waste

FoodLoop AI: Demand Prediction → Waste Prevention → Smart Matching → Safe Redistribution → Impact Analytics

FOR ADMINISTRATORS / INSTITUTIONS

- Centralized food waste monitoring
- AI-driven operational insights
- Institution-wise waste analytics
- Track redistribution & sustainability metrics
- Scalable across multiple kitchens and locations

FOR KITCHEN MANAGERS

- Identify high-waste menu items
- Detect recurring waste patterns
- Optimize preparation quantities
- Get AI-based production recommendations
- Monitor kitchen performance

RESEARCH AREAS

- AI-Based Food Waste Prediction
- Demand Forecasting for Institutional Kitchens
- Food Waste Classification & Analytics
- Smart Food Redistribution Systems
- AI-Based Matching & Route Optimization
- Food Safety & Shelf-Life Prediction
- Computer Vision for Food Waste Detection
- Sustainable Supply Chain & Circular Economy

REFERENCES

- UNEP – Food Waste Index Report
- FAO – Food Loss and Food Waste
- UN Sustainable Development Goals – SDG 2 & SDG 12
- WRAP – Food Waste Reduction & Redistribution
- FAO – Sustainable Food Systems
- Research on AI-Based Food Waste Prediction & Demand Forecasting
- Research on Computer Vision for Food Waste Classification
- Research on Food Rescue & Redistribution Optimization